

GraphQL Blog — Project Overview

Project: GraphQL Blog

Technology: Python, FastAPI, Strawberry GraphQL, SQLAlchemy, PostgreSQL

Location: /home/dominic/Documents/graphql-blog

Report #: 1/20

Overview

GraphQL-powered blog with Strawberry, SQLAlchemy, PostgreSQL, and full CRUD.

Purpose

Serves as a GraphQL-based blog platform with user management and post CRUD.

Status

Active development. See report 16 (Known Issues) and report 19 (Improvements).

Key Features

• GraphQL API with Strawberry • User auth via JWT • Post CRUD with pagination • PostgreSQL database

Entry Points

`docker compose up` or via daemon

Detailed Architecture

Project Structure

`app/main.py` (FastAPI + Strawberry), `app/schema.py` (GraphQL types + resolvers), `app/models.py`, `app`

Technology Stack

Layer	Technology	Purpose
Backend Framework	FastAPI	REST/API server

Layer	Technology	Purpose
Database	SQLite	Data persistence
Auth	JWT	Authentication & authorization
Frontend	None	User interface
Deployment	Render.com + GitHub Pages	Hosting & CI/CD

Module Breakdown

See project structure above.

Data Flow

1. Request enters via HTTP (REST, GraphQL, or WebSocket)
2. Authentication middleware validates JWT token (if required)
3. Router dispatches to appropriate handler
4. Handler processes business logic with database queries
5. Response formatted and returned to client
6. Frontend renders response in appropriate view

Entry Points

- **Server:** `uvicorn app.main:app`
- **Port:** 8000 (default)
- **Health check:** `GET /health`

Configuration

Environment Variables

Variable	Description	Default
DATABASE_URL	Database connection string	sqlite:///./data.db
SECRET_KEY	JWT signing secret	(required)
CORS_ORIGINS	Allowed CORS origins	*
DEBUG	Debug mode flag	false

Deployment

Requirements

- Python 3.10+
- pip dependencies (see requirements.txt)
- SQLite (or PostgreSQL for production)

Startup Commands

```
# Development
uvicorn app.main:app --reload

# Production
uvicorn app.main:app --host 0.0.0.0 --port 8000
```

Related Reports

- Report 02: Architecture Analysis (detailed component diagram)
- Report 05: Tech Stack & Dependencies (version-specific)
- Report 13: Deployment Guide (Docker, CI/CD)